

GEMM phase: one CTA for each $i \leq j$

$q = 4$ output-tile rows and columns

row i	0	column j	2	3
0	(0,0)	(0,1)	(v ,0)	(3,1)
1		(1,0)	(1,1)	(v ,1)
2			(2,0)	(2,1)
3				(3,0)

cell label = (s, d) ; virtual start $v = q = 4$

Completed symmetric matrix

computed upper tiles and mirrored lower tiles

row i	0	column j	2	3
0	C_{00}	C_{01}	C_{02}	C_{03}
1	C_{01}^T	C_{11}	C_{12}	C_{13}
2	C_{02}^T	C_{12}^T	C_{22}	C_{23}
3	C_{03}^T	C_{13}^T	C_{23}^T	C_{33}

mirror lower tiles
by shared-memory transpose¹



$$C_{ji} \leftarrow C_{ij}^T \\ \text{for } j > i$$

light: GEMM output

dark: transpose copy